Context:

- Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.

- This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.

- By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

Problem Statement:

● Extracting valuable insights and providing actionable recommendations from the given data

Data Description:

The data is available in 8 different csv files:

1. customers.csv
2. geolocation.csv
3. order_items.csv
4. payments.csv
5. reviews.csv
6. orders.csv
7. products.csv
8. Sellers.csv

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Analysis

I. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

```
SELECT
  column_name,
  data_type
FROM
  target.INFORMATION_SCHEMA.COLUMNS
WHERE
  table_name = 'customers'
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

* INFORMATION_SCHEMA.COLUMNS is a special function in bigquery.

Insights:

- In the customers table, we have 5 columns, in which 4 are of String data type and 1 is integer.

2. Get the time range between which the orders were placed.

```
SELECT
  MIN(order_purchase_timestamp) AS earliest_order_date,
  MAX(order_purchase_timestamp) AS latest_order_date,
  DATE_DIFF(
    DATE(MAX(order_purchase_timestamp)),
    DATE(MIN(order_purchase_timestamp)),
    DAY
  ) AS total_days_range
FROM target.orders;
```

Row	earliest_order_date	latest_order_date	total_days_range
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	773

Insight:

- This query shows the earliest and latest order dates, giving the full time range in which customers placed orders.

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3.Count the Cities & States of customers who ordered during the given period.

```
SELECT
  g.geolocation_city AS city,
  g.geolocation_state AS state,
  COUNT(*) AS total_customers
FROM
  target.orders o
JOIN
  target.customers c ON c.customer_id = o.customer_id
JOIN
  target.geolocation g ON c.customer_zip_code_prefix =
g.geolocation_zip_code_prefix
GROUP BY
  city,
  state
ORDER BY
  total_customers DESC;
```

Row	city	state	total_customers
1	rio de janeiro	RJ	1913894
2	sao paulo	SP	1164450
3	belo horizonte	MG	737556
4	niteroi	RJ	393175
5	curitiba	PR	255731

Insight:

- This query joins orders, customers and geolocation data to count how many customers ordered from each city and state.

2.In-depth Exploration:

1.Is there a growing trend in the no. of orders placed over the past years?

```
from target.orders;
WITH yearly_orders AS(
  SELECT
  EXTRACT(year  FROM  order_delivered_customer_date) AS order_year,
  COUNT(o.order_id) AS total_orders
  FROM target.orders o
  GROUP BY 1
)
SELECT
  order_year,
```

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```
  total_orders,
  lag(total_orders) OVER (order by order_year) AS prev_year_orders,total_orders -
LAG(total_orders) OVER(ORDER BY order_year) AS order_growth
FROM
  yearly_orders
ORDER BY
  order_year;
```

Row	order_year	total_orders	prev_year_orders	order_growth
1	null	2965	null	null
2	2016	272	2965	-2693
3	2017	40930	272	40658
4	2018	55274	40930	14344

Insight:

Orders shows a clear year on year increase from 2016 to 2018, indicating a growing trend

2.Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
WITH t AS (
SELECT
  FORMAT_TIMESTAMP ('%B-%Y', order_purchase_timestamp) AS month_seasonality,
  EXTRACT (year FROM order_purchase_timestamp) AS yr,
  EXTRACT (month FROM order_purchase_timestamp) AS mn,
  COUNT(order_id) AS total_orders
FROM
  target. orders
GROUP BY
 month_seasonality, yr, mn
)
SELECT
  month_seasonality,
  total_orders
FROM
  t
ORDER BY
  yr, mn;
```

Row	month_seasonality	total_orders
1	September-2016	4
2	October-2016	324
3	December-2016	1
4	January-2017	800
5	February-2017	1780
6	March-2017	2682
7	April-2017	2404

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Insight:
The query reveals a clear upward trend in month orders peaking in January 2017.

3.During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs: Dawn
- 7-12 hrs: Mornings
- 13-18 hrs: Afternoon
- 19-23 hrs: Night

```
SELECT
  CASE
  when EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6
THEN 'Dawn'
  when EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12
THEN 'Mornings'
  when EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18
THEN 'Afternoon'
  when EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 19 AND 23
THEN 'Night'
  END AS order_category,
  COUNT(order_id) AS order_count
FROM
  target. orders
GROUP BY
  order_category
ORDER BY
  order_count DESC;
```

Row	order_purchase_timestamp	time_of_the_day	order_category
1	2017-02-01 17:31:17 UTC	1	Afternoon
2	2017-03-01 11:26:37 UTC	1	Mornings
3	2017-05-01 16:12:39 UTC	1	Afternoon
4	2017-09-01 22:43:44 UTC	1	Night
5	2017-10-01 08:15:15 UTC	1	Mornings
6	2017-10-01 22:26:25 UTC	1	Night

Insight:
Brazilian customers primarily place orders in the evening (6 PM to 10 PM) and afternoon (1 PM to 4 PM). They tend to shop most during the week.

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3.Evolution of E-commerce orders in the Brazil region:

1.Get the month on month no. of orders placed in each state.

```
SELECT
  c.customer_state,
  FORMAT_TIMESTAMP ('%B', o. order_purchase_timestamp) AS month_name,
  EXTRACT (month from order_purchase_timestamp) AS month_number,
  COUNT (o. order_id) AS num_orders
FROM
  target. orders o
JOIN
  target. customers c ON c. customer_id = o. customer_id
GROUP BY
  customer_state,
  month_name,
  month_number
ORDER BY
  customer_state,
  month_number ASC;
```

Row	customer_state	month_name	month_number	num_orders
1	AC	January	1	8
2	AC	February	2	6
3	AC	March	3	4
4	AC	April	4	9
5	AC	May	5	10
6	AC	June	6	7

Insight:
RR has the most consistent orders across months, while other states like AP and AM show orders only in specific months.

2.How are the customers distributed across all the states?

```
SELECT
  customer_state,
  COUNT(customer_id) AS num_customers
FROM
  target. customers
GROUP BY
  customer_state
ORDER BY
  num_customers DESC;
```

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Row	customer_state	num_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045

Insight:
Most customers come from a few major states, while many states have very few customers.

4.Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

```
WITH yearly_cost AS (
  SELECT
  EXTRACT (year from order_purchase_timestamp) AS year,
  SUM (p. payment_value) AS total_cost
  FROM target. orders o
  JOIN target. payments p ON p.order_id = o.order_id
  WHERE EXTRACT (month from order_purchase_timestamp) BETWEEN 1 AND 8
AND
  EXTRACT (year from order_purchase_timestamp) IN (2017, 2018)
  GROUP BY year
),
calc AS (
  SELECT
  MAX (CASE WHEN year = 2017 THEN total_cost END) AS cost_2017,
  MIN (CASE WHEN year = 2018 THEN total_cost END) AS cost_2018
  FROM yearly_cost
)
SELECT
  cost_2017,
  cost_2018,
  ROUND (((cost_2018 - cost_2017) / cost_2017) * 100, 2) AS percentage_increase
FROM
   calc;
```

Row	cost_2017	cost_2018	percentage_incre...
1	3669022.119999...	8694733.840000...	136.98

Insight:
Increase marketing and promotional activities during high-spending months in 2018

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to further boost customer purchases and maximize revenue growth.

2.Calculate the Total & Average value of order price for each state.

```
SELECT
  c.customer_state,
  ROUND (SUM (p. payment_value), 2) AS total_order_palced,
  ROUND (AVG (p. payment_value), 2) AS avg_order_price
FROM
  target. customers c
JOIN
  target. orders o ON c. customer_id = o. customer_id
JOIN
  target. payments p ON o. order_id = p. order_id
GROUP BY
  c.customer_state;
```

Row	customer_state	total_order_palced	avg_order_price
1	AC	19680.62	234.29
2	AL	96962.06	227.08
3	AM	27966.93	181.6
4	AP	16262.8	232.33
5	BA	616645.82	170.82

Insight:
The average order price varies significantly across differently states, indicating potential regional differences in customer spending habits.

3.Calculate the Total & Average value of order freight for each state.

```
SELECT
   c.customer_state,
   ROUND (SUM (oi. freight_value), 2) AS total_freight_value,
   ROUND (AVG (oi. freight_value), 2) AS average_freight_value
FROM
   target. order_items oi
JOIN
   target. orders o ON oi. order_id = o. order_id
JOIN
   target. customers c ON o. customer_id = c. customer_id
GROUP BY
   c.customer_state
ORDER BY
   c.customer_state;
```

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Row	customer_state	total_freight_value	average_freight_v...
1	AC	3686.75	40.07
2	AL	15914.59	35.84
3	AM	5478.89	33.21
4	AP	2788.5	34.01
5	BA	100156.68	26.36

Insight:
States with high total freight have many orders, high average freight indicates costly deliveries, and low total & average shows opportunities for promotion.

5.Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
   Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
   Do this in a single query.

   You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:
   - time_to_deliver = order_delivered_customer_date - order_purchase_timestamp
   - diff_estimated_delivery = order_delivered_customer_date - order_estimated_delivery_date

```
SELECT
    DATE_DIFF (order_delivered_customer_date, order_purchase_timestamp, day)
AS          time_to_delivery,
    DATE_DIFF (order_delivered_customer_date, order_estimated_delivery_date, day)
AS   diff_estimated_delivery
FROM
    target. orders;
```

Row	time_to_delivery	diff_estimated_d...
1	35	-16
2	30	-28
3	30	12
4	54	36
5	43	-6

Insight:

- Most orders take around 30–40 days to deliver.

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- Many orders are delivered earlier than estimated (negative values).
- Some orders are delivered later than estimated (positive values)
- Overall, delivery performance is mostly on time or early with few delays.

2.Find out the top 5 states with the highest & lowest average freight value.

```
WITH freight_value AS (
  SELECT
  c.customer_state,
  AVG (oi. freight_value) AS avg_freight_value
  FROM
  target. order_items oi
  JOIN
  target. orders o ON oi. order_id = o. order_id
  JOIN
  target. customers c ON o. customer_id = c. customer_id
  GROUP BY
  c.customer_state
)
(
SELECT
  customer_state,
  ROUND (avg_freight_value, 2) AS avg_freight_value, 'Highest' AS category
FROM
  freight_value
ORDER BY
  avg_freight_value DESC
LIMIT
  5
)

UNION ALL

SELECT
  customer_state,
  ROUND (avg_freight_value, 2) AS avg_freight_value, 'Lowest' AS category
FROM
  freight_value
ORDER BY
  avg_freight_value ASC
LIMIT
  5;
```

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Row	customer_state	avg_freight_value	category
1	SP	15.15	Lowest
2	PR	20.53	Lowest
3	MG	20.63	Lowest
4	RJ	20.96	Lowest
5	DF	21.04	Lowest

3.Find out the top 5 states with the highest & lowest average delivery time.

```
WITH StateAvgDurations AS (

  SELECT
    c.customer_state,
    AVG (TIMESTAMP_DIFF(o.order_delivered_customer_date,
o.order_purchase_timestamp, SECOND)) AS avg_delivery_seconds
  FROM

    `target. orders` AS o
  JOIN
    `target. customers` AS c ON o. customer_id = c. customer_id
  WHERE
    o. order_delivered_customer_date IS NOT NULL -- Exclude undelivered orders
  GROUP BY
    c.customer_state
),
RankedDurations AS (

  SELECT
    customer_state,
    avg_delivery_seconds,
    -- Convert seconds to hours and round for display
    ROUND (avg_delivery_seconds / 3600, 2) AS avg_delivery_hours,
    RANK () OVER (ORDER BY avg_delivery_seconds DESC) as rank_desc,
    RANK () OVER (ORDER BY avg_delivery_seconds ASC) as rank_asc
  FROM
    StateAvgDurations
)

SELECT
  customer_state,
```

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```
    avg_delivery_hours,
    'Highest' AS category
FROM
    RankedDurations
WHERE
    rank_desc <= 5

UNION ALL


SELECT
    customer_state,
    avg_delivery_hours,
    'Lowest' AS category
FROM
    RankedDurations
WHERE
    rank_asc <= 5
ORDER BY
   avg_delivery_hours DESC;
```

Row	customer_state	avg_delivery_hours	category
1	RR	705.3	Highest
2	AP	652.44	Highest
3	AM	634.22	Highest
4	AL	589.05	Highest
5	PA	570.55	Highest
6	SC	359.02	Lowest
7	DF	311.22	Lowest

4.Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.
You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```
SELECT
    c.customer_state AS state,
    ROUND (AVG (DATE_DIFF (o. order_estimated_delivery_date,
                    o. order_delivered_customer_date,
                    DAY)), 2) AS avg_days_early
FROM
     target. orders o
JOIN
    target. customers c
    ON o. customer_id = c. customer_id
```

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```
WHERE
    o. order_delivered_customer_date IS NOT NULL
  AND o. order_estimated_delivery_date IS NOT NULL
GROUP BY
      state
ORDER BY
  avg_days_early DESC
LIMIT 5;
```

Row	state	avg_days_early
1	AC	19.76
2	RO	19.13
3	AP	18.73
4	AM	18.61
5	RR	16.41

Insight:
Orders in AC, RO, AP, AM, and RR are delivered the earliest, showing these states have the fastest and most efficient delivery performance.

6.Analysis based on the payments:

1.Find the month on month no. of orders placed using different payment types.

```
SELECT
   DATE_TRUNC (order_approved_at, MONTH) AS month,
   payment_type,
   COUNT (DISTINCT o. order_id) AS total_orders
FROM
   target. orders o
JOIN
   target. payments p
ON
   o. order_id = p. order_id
WHERE
   order_approved_at IS NOT NULL
GROUP BY
   month, payment_type
ORDER BY
   month, payment_type;
```

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Row	month	payment_type	total_orders
1	2016-10-01 00:00:00 UTC	UPI	61
2	2016-10-01 00:00:00 UTC	credit_card	252
3	2016-10-01 00:00:00 UTC	debit_card	2
4	2016-10-01 00:00:00 UTC	voucher	10
5	2016-12-01 00:00:00 UTC	credit_card	1

Insight:

- Credit card is usually the most used payment method every month.
- Bolete and debit card show lower and stable usage.
- Peaks in orders usually match holiday months (Nov–Dec).

2.Find the no. of orders placed on the basis of the payment instalments that have been paid.

```
SELECT
   payment_installments,
   COUNT (DISTINCT order_id) AS total_orders
FROM
   target.payments
GROUP BY
   payment_installments
ORDER BY
   payment_installments;
```

Row	payment_installm...	total_orders
1	0	2
2	1	49060
3	2	12389
4	3	10443
5	4	7088

Insight:

- Most customers choose 1 instalment → indicates preference for full payment.
- Instalments like 2–6 are used but much less.
- Very high instalment counts (10–12) are rare, showing limited long-term ffinancing usage.

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Actionable Insights & Recommendations

Based on the full analysis of orders, customer behaviour, delivery performance, freight patterns, and payment methods across Brazil, the following insights highlight clear opportunities for improving business efficiency, customer satisfaction, and revenue growth.

📌 1. Improve Delivery Speed in High-Delay States

Some states show significantly longer delivery times.

Recommendations:

- Partner with faster regional carriers
- Add local fulfilment/warehouse points
- Improve route planning for remote states

📌 2. Reduce High Freight Costs

States with higher average freight values increase overall order expenses.

Recommendations:

- Negotiate better rates with courier partners
- Promote free shipping thresholds for customers
- Encourage sellers to stock products closer to buyers

📌 3. Expand Market Presence in Low-Order States

Many states show fewer customers and very few orders.

Recommendations:

- Run targeted regional marketing campaigns

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- Offer location-based discounts to attract new customers
- Improve delivery accessibility in underserved areas

📌 4. Use Seasonal & Monthly Trends to Increase Revenue

Order volume shows clear seasonal variation.

Recommendations:

- Launch promotional offers during high-demand months
- Increase inventory ahead of peak seasons
- Offer holiday discounts, bundles, and flash sales

📌 5. Leverage Payment Behaviour to Boost Sales

Most customers use credit card and prefer single-instalment payments.

Recommendations:

- Promote zero-interest EMI for high-value purchases
- Increase cashback offers for popular payment methods
- Partner with banks for co-branded credit promotions

📌 6. Improve Delivery Estimate Accuracy

Many orders are delivered earlier or later than predicted.

Recommendations:

- Use predictive models for more accurate delivery estimates
- Update estimated delivery times based on state-level history

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- Notify customers proactively about early or delayed deliveries

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📌 7. Strengthen High-Value Customer Regions

Some states consistently show higher order value and spending.

Recommendations:

- Launch loyalty/points-based rewards
- Offer exclusive deals for premium customers
- Promote subscription-based free/fast delivery options

✨ Final Summary

This analysis provides a complete understanding of customer ordering behaviour, payment preferences, state-wise performance, freight impact, and delivery efficiency in Brazil. The insights reveal strong growth trends, clear seasonality, and important regional differences. By applying the recommended strategies—such as improving logistics, optimizing freight, targeting low-order states, and enhancing payment offers—the business can significantly improve operational efficiency, customer satisfaction, and overall revenue growth.

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